

LP-V Practical 1: OpenMP Parallel BFS & DFS - Viva Q&A;

LP-V (410255) Practical Exam Viva Question Bank with Answers

OpenMP BFS/DFS Viva Questions

Q1. What is BFS?

Ans: BFS (Breadth First Search) visits nodes level-by-level using a queue.

Q2. What is DFS?

Ans: DFS (Depth First Search) explores depth-wise using recursion/stack.

Q3. Why BFS is suitable for parallelization?

Ans: Because all nodes in same level can be processed independently in parallel.

Q4. Why DFS is difficult to parallelize?

Ans: DFS has recursive dependency and sequential traversal order constraints.

Q5. What data structure is used in BFS?

Ans: Queue.

Q6. What data structure is used in DFS?

Ans: Stack/Recursion.

Q7. How to parallelize BFS using OpenMP?

Ans: Use parallel for to process adjacency list of all nodes at current frontier.

Q8. What is race condition in BFS/DFS parallel code?

Ans: When multiple threads update visited array/queue simultaneously causing incorrect results.

Q9. How to avoid race condition?

Ans: Use critical/atomic section while updating shared variables.

Q10. What is OpenMP directive used for BFS/DFS loop?

Ans: #pragma omp parallel for

Q11. What is barrier in OpenMP?

Ans: Barrier makes all threads wait until all reach synchronization point.

Q12. What is critical section?

Ans: Ensures only one thread executes that code block at a time.

Q13. What is atomic in OpenMP?

Ans: Ensures atomic update of a shared variable.

Q14. Explain speedup formula.

Ans: $\text{Speedup} = T_{\text{serial}} / T_{\text{parallel}}$.

Q15. What is efficiency?

Ans: $\text{Efficiency} = \text{Speedup} / \text{number of processors}$.

LP-V Practical 2: OpenMP Parallel Sorting (Bubble & Merge)

- Viva Q&A;

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OpenMP Sorting Viva Questions

Q1. What is Bubble Sort?

Ans: It repeatedly swaps adjacent elements if they are in wrong order.

Q2. Time complexity of Bubble Sort?

Ans: $O(n^2)$.

Q3. Why bubble sort is inefficient in parallel?

Ans: Because each pass depends on previous swaps causing dependency.

Q4. What is Merge Sort?

Ans: Divide and conquer algorithm: divide array, sort subarrays, merge.

Q5. Time complexity of Merge Sort?

Ans: $O(n \log n)$.

Q6. Why merge sort is good for parallel execution?

Ans: Subarrays can be sorted independently in parallel.

Q7. How to parallelize merge sort using OpenMP?

Ans: Use parallel sections or tasks for recursive sorting.

Q8. What is the merge step?

Ans: Combining two sorted arrays into one sorted array.

Q9. Define speedup in sorting experiment.

Ans: $\text{Speedup} = \text{sequential time} / \text{parallel time}$.

Q10. What is scalability?

Ans: Ability of algorithm to improve performance when threads increase.

Q11. What is load balancing in sorting?

Ans: Distributing equal work among threads to avoid idle threads.

Q12. What is the use of `omp_get_wtime()`?

Ans: To measure execution time in OpenMP programs.

Q13. What is the difference between static and dynamic scheduling?

Ans: Static divides iterations beforehand; dynamic assigns during runtime.

Q14. Which sorting is stable?

Ans: Merge sort is stable; bubble sort is stable too.

Q15. Which sorting is faster for large n ?

Ans: Merge sort.

LP-V Practical 3: OpenMP Reduction (Min/Max/Sum/Avg) - Viva Q&A;

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OpenMP Reduction Viva Questions

Q1. What is reduction in parallel computing?

Ans: Combining partial results into final result like sum/min/max.

Q2. OpenMP syntax for sum reduction?

Ans: `#pragma omp parallel for reduction(+:sum)`

Q3. How to compute max using reduction?

Ans: `#pragma omp parallel for reduction(max:max_val)`

Q4. How to compute min using reduction?

Ans: `#pragma omp parallel for reduction(min:min_val)`

Q5. How to compute average using reduction?

Ans: Average = sum/n after reduction.

Q6. Why reduction is faster than using critical section?

Ans: Reduction avoids heavy synchronization and uses private partial sums.

Q7. What is shared variable?

Ans: One copy accessible to all threads.

Q8. What is private variable?

Ans: Separate copy for each thread.

Q9. What is false sharing?

Ans: Performance issue when threads modify variables in same cache line.

Q10. What is race condition in sum calculation?

Ans: Multiple threads update sum simultaneously causing wrong result.

Q11. How reduction avoids race condition?

Ans: Each thread uses local sum then combines at end.

Q12. What is barrier at end of reduction?

Ans: Ensures all threads finish before combining results.

Q13. What is the complexity of reduction sum?

Ans: $O(n)$ work but parallel reduces time to $O(n/p)$.

Q14. What is OpenMP environment variable OMP_NUM_THREADS?

Ans: Used to set number of threads.

Q15. What is the use of `omp_set_num_threads()`?

Ans: Programmatically set number of threads.

LP-V Practical 4: CUDA Vector Addition - Viva Q&A;

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CUDA Vector Addition Viva Questions

Q1. What is CUDA?

Ans: NVIDIA parallel computing platform to run code on GPU.

Q2. What is kernel in CUDA?

Ans: A function executed on GPU by many threads.

Q3. How to declare kernel?

Ans: Using `__global__` keyword.

Q4. Difference between host and device memory?

Ans: Host = CPU RAM, Device = GPU global memory.

Q5. What is cudaMalloc?

Ans: Allocates memory on GPU.

Q6. What is cudaMemcpy?

Ans: Copies data between host and device.

Q7. What is cudaFree?

Ans: Frees GPU memory.

Q8. Explain vector addition in CUDA.

Ans: Each thread computes $C[i] = A[i] + B[i]$.

Q9. Formula for global thread index?

Ans: $i = blockDim.x * blockIdx.x + threadIdx.x$

Q10. What is blockDim?

Ans: Number of threads in a block.

Q11. What is gridDim?

Ans: Number of blocks in a grid.

Q12. Why GPU is faster for vector addition?

Ans: Thousands of threads execute operations in parallel.

Q13. What is thread divergence?

Ans: When threads in same warp follow different execution paths.

Q14. What is warp?

Ans: Group of 32 threads executed together in GPU.

Q15. What is kernel launch syntax?

Ans: `kernel<<>>(args);`

LP-V Practical 5: CUDA Matrix Multiplication - Viva Q&A;

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CUDA Matrix Multiplication Viva Questions

Q1. What is matrix multiplication?

Ans: $C[i][j] = \text{sum}(A[i][k]*B[k][j])$.

Q2. How matrix multiplication is parallelized in CUDA?

Ans: Each thread computes one element of output matrix.

Q3. Time complexity of matrix multiplication?

Ans: $O(n^3)$.

Q4. What is shared memory in CUDA?

Ans: Fast memory shared among threads in same block.

Q5. Why shared memory improves matrix multiplication?

Ans: Reduces global memory access by storing tiles locally.

Q6. What is tiling technique?

Ans: Divide matrix into tiles processed using shared memory.

Q7. What is global memory?

Ans: Main GPU memory accessible by all blocks but slower.

Q8. What is coalesced memory access?

Ans: When consecutive threads access consecutive memory locations for efficiency.

Q9. Explain threadIdx.x and threadIdx.y usage.

Ans: Used for 2D indexing in matrix multiplication.

Q10. What is blockIdx.x and blockIdx.y?

Ans: Block position in grid for 2D blocks.

Q11. How to define 2D grid in CUDA?

Ans: `dim3 grid(xBlocks, yBlocks); dim3 block(tx, ty);`

Q12. What is cudaDeviceSynchronize?

Ans: Waits until all GPU threads finish execution.

Q13. What is the output of matrix multiplication?

Ans: A new matrix representing product of input matrices.

Q14. What is occupancy in GPU?

Ans: How many threads are active on GPU relative to maximum.

Q15. Why choose block size 16x16 or 32x32?

Ans: It matches warp execution and improves memory efficiency.

LP-V Practical 6: MPI Mini Project (Parallel Quicksort) - Viva Q&A;

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MPI Viva Questions

Q1. What is MPI?

Ans: Message Passing Interface used for distributed memory parallel computing.

Q2. Difference between MPI and OpenMP?

Ans: MPI uses multiple processes with message passing; OpenMP uses threads with shared memory.

Q3. What is MPI_Send?

Ans: Sends message/data from one process to another.

Q4. What is MPI_Recv?

Ans: Receives message/data from another process.

Q5. What is MPI_Bcast?

Ans: Broadcasts data from root process to all processes.

Q6. What is MPI_Reduce?

Ans: Combines values from all processes and returns result to root.

Q7. What is rank in MPI?

Ans: Unique ID of each process.

Q8. What is communicator in MPI?

Ans: Group of processes that can communicate (MPI_COMM_WORLD).

Q9. What is parallel quicksort idea?

Ans: Divide array among processes, sort locally, exchange partitions and merge.

Q10. What is load balancing in MPI quicksort?

Ans: Ensuring equal data distribution among processes.

Q11. What is communication overhead?

Ans: Time spent in sending/receiving messages between processes.

Q12. What is deadlock in MPI?

Ans: When processes wait indefinitely for each other's messages.

Q13. How to avoid deadlock?

Ans: Use proper send/receive order or non-blocking communication.

Q14. What is MPI_Isend and MPI_Irecv?

Ans: Non-blocking send/receive functions.

Q15. What is advantage of MPI?

Ans: Works on clusters and distributed systems.

LP-V DL Practical 1: Boston Housing Regression using DNN - Viva Q&A;

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Boston Housing DNN Viva Questions

Q1. What is regression?

Ans: Regression predicts continuous values such as house prices.

Q2. Why use DNN for Boston housing dataset?

Ans: To learn nonlinear relationship between features and house price.

Q3. Which activation is used in hidden layers?

Ans: ReLU is commonly used.

Q4. Which activation is used in output layer for regression?

Ans: Linear activation (no activation) is used.

Q5. Which loss function is used for regression?

Ans: Mean Squared Error (MSE).

Q6. Define MSE.

Ans: $MSE = (1/n) \sum (y - y_{pred})^2$

Q7. Define RMSE.

Ans: $RMSE = \sqrt{MSE}$.

Q8. Why normalize input features?

Ans: To bring all features to same scale for faster convergence.

Q9. What optimizer is commonly used?

Ans: Adam optimizer.

Q10. What is overfitting?

Ans: Model performs well on training data but poorly on test data.

Q11. How to reduce overfitting?

Ans: Dropout, early stopping, regularization, more data.

Q12. What is validation split?

Ans: A part of training data used for validation during training.

Q13. Why use train-test split?

Ans: To evaluate model on unseen data.

Q14. What is epoch?

Ans: One complete pass of dataset through model.

Q15. What is batch size?

Ans: Number of samples processed before updating weights.

LP-V DL Practical 2: Classification using Deep Neural Network - Viva Q&A;

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DNN Classification Viva Questions

Q1. What is classification?

Ans: Predicting discrete class labels.

Q2. Difference between binary and multiclass classification?

Ans: Binary has 2 classes, multiclass has more than 2 classes.

Q3. Activation for binary classification output layer?

Ans: Sigmoid.

Q4. Activation for multiclass classification output layer?

Ans: Softmax.

Q5. Loss function for binary classification?

Ans: Binary crossentropy.

Q6. Loss function for multiclass classification?

Ans: Categorical crossentropy.

Q7. What is one-hot encoding?

Ans: Converts class label into vector of 0s and 1s.

Q8. Define accuracy.

Ans: $\text{Accuracy} = (\text{correct predictions}) / (\text{total predictions})$.

Q9. Define precision.

Ans: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$.

Q10. Define recall.

Ans: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$.

Q11. Define F1-score.

Ans: $\text{F1} = 2\text{PR} / (\text{P} + \text{R})$.

Q12. What is confusion matrix?

Ans: Matrix showing TP, TN, FP, FN counts.

Q13. What is ROC curve?

Ans: Graph between TPR and FPR.

Q14. What is AUC?

Ans: Area under ROC curve; higher is better.

Q15. Why use embedding in text classification?

Ans: To convert words into dense vectors.

Q16. Why use dropout in classification model?

Ans: To prevent overfitting.

LP-V DL Practical 3: CNN (Fashion MNIST / Plant Disease) - Viva Q&A;

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CNN Viva Questions

Q1. What is CNN?

Ans: Convolutional Neural Network used for image processing.

Q2. Why CNN is better than ANN for images?

Ans: It captures spatial features and reduces parameters.

Q3. What is convolution layer?

Ans: Applies filters to input image to generate feature maps.

Q4. What is kernel/filter?

Ans: Small matrix (3x3, 5x5) used to detect features.

Q5. What is feature map?

Ans: Output of convolution showing extracted features.

Q6. What is stride?

Ans: Step size by which kernel moves.

Q7. What is padding?

Ans: Adding extra pixels to preserve spatial size.

Q8. What is pooling layer?

Ans: Downsamples feature map to reduce size.

Q9. What is max pooling?

Ans: Selects maximum value from region.

Q10. What is flatten layer?

Ans: Converts 2D feature map to 1D vector.

Q11. Why dropout used in CNN?

Ans: To reduce overfitting.

Q12. What is batch normalization?

Ans: Normalizes activations to improve training stability.

Q13. Explain Fashion MNIST dataset.

Ans: Dataset of 28x28 grayscale images of 10 clothing classes.

Q14. What is typical CNN architecture?

Ans: Conv -> ReLU -> Pool -> Conv -> Pool -> Flatten -> Dense -> Softmax.

Q15. How to improve CNN accuracy?

Ans: Increase data, augmentation, deeper network, tuning learning rate.

LP-V DL Practical 4: RNN/LSTM Google Stock Price Prediction - Viva Q&A;

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RNN/LSTM Viva Questions

Q1. What is RNN?

Ans: Recurrent Neural Network used for sequential/time series data.

Q2. Why RNN is used for stock prediction?

Ans: Because stock prices depend on previous time steps.

Q3. What is LSTM?

Ans: Long Short-Term Memory network, improved RNN with gates.

Q4. Why LSTM preferred over RNN?

Ans: LSTM handles long-term dependencies and avoids vanishing gradients.

Q5. What is vanishing gradient problem?

Ans: Gradients become very small, preventing learning long sequences.

Q6. What are timesteps in RNN?

Ans: Number of past days used to predict future value.

Q7. Why scaling is required for stock data?

Ans: To bring values into range for stable training.

Q8. Which loss function is used for stock prediction?

Ans: MSE is commonly used.

Q9. Which activation is used in LSTM layer?

Ans: tanh and sigmoid gates internally.

Q10. What is sequence-to-one model?

Ans: Uses input sequence to predict one output value.

Q11. What is train-test split in time series?

Ans: Split based on time order, not random.

Q12. Why shuffle is false in time series?

Ans: Because order matters for learning sequence.

Q13. What is sliding window method?

Ans: Create sequences using window of past values.

Q14. What is prediction output in stock model?

Ans: Next day price or future price.

Q15. How to improve stock prediction?

Ans: More data, better features, tuning LSTM units, using GRU.